

BCAN Progress Report



Team Members

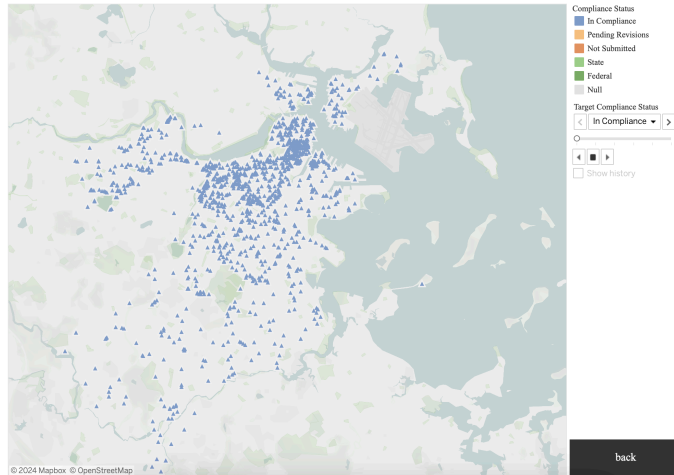
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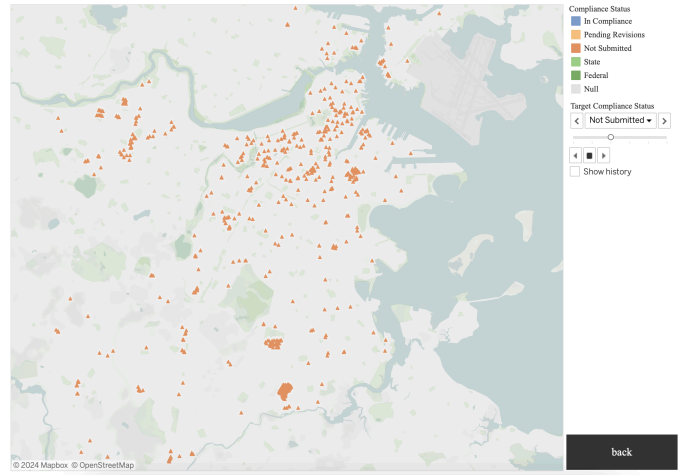
Portfolio Overview

Project Overview: As director of advocacy, I would like an overview of all the BERDO-required buildings in Boston so that I can quickly spot compliant and deficient buildings.

Overview of BERDO-required Buildings Compliance Status



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Overview of BERDO-required Buildings Compliance Status

Key Insights:

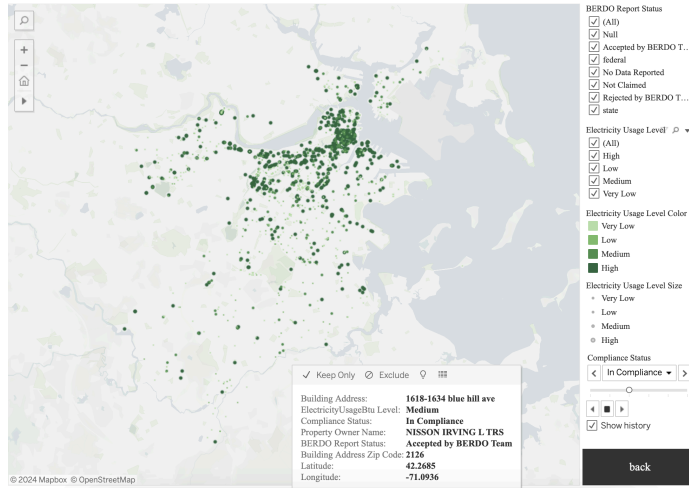
1. Compliance Distribution:
 - a. Buildings categorized as “In Compliance” are clustered in central Boston, reflecting strong adherence to BERDO standards in commercial districts.
 - b. Non-compliance cases (“Not Submitted”) are more dispersed, particularly in residential and suburban regions, highlighting areas that need focused engagement.
2. Geographic Trends:
 - a. Emission hotspots are evident in areas with older, larger buildings, particularly in institutional and industrial zones.
 - b. Residential neighborhoods in lower-income zip codes display higher rates of non-compliance, possibly linked to socioeconomic disparities.
3. Interactive Capabilities:
 - a. Filters allow stakeholders to explore compliance statuses by building type, zip code, and other attributes.
 - b. Dynamic visualizations enhance the ability to pinpoint specific neighborhoods or building categories needing immediate action.

Policy Recommendations:

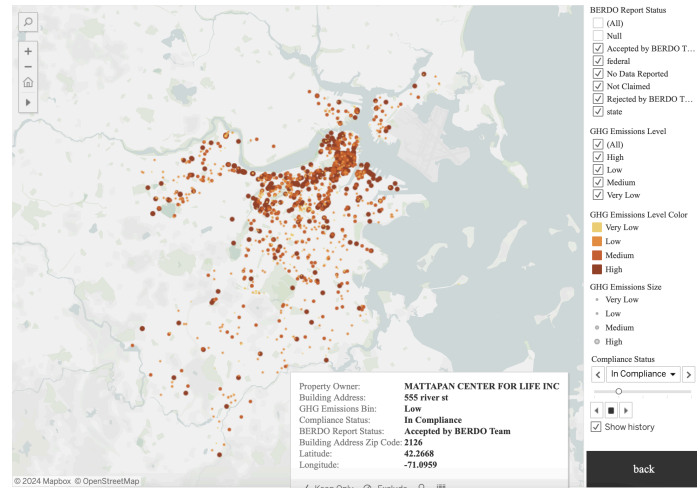
1. Targeted Outreach:
 - a. Focus advocacy efforts on neighborhoods with high non-compliance rates and older building stock.

- b. Prioritize engagement with property owners of institutional and industrial buildings.
 2. Enhanced Visualization Tools:
 - a. Develop additional geospatial layers to correlate compliance with socio-economic factors, such as income and energy use intensity.
 - b. Expand the use of historical data to track compliance trends over time and measure the effectiveness of interventions.
 3. Broader Participation in Renewable Energy:
 - a. Promote renewable energy initiatives, such as Power Purchase Agreements (PPAs), in neighborhoods with lower compliance rates.
 - b. Tailor incentives to property types and owners who face significant operational or financial challenges.

Overview of BERDO-required Buildings in Boston with Electricity Usage



Overview of BERDO-required Buildings in Boston with GHG Emissions



GHG Emissions and Electricity Usage of BERDO-Required Buildings in Boston

Key Insights:

1. GHG Emissions by Building:
 - a. The map highlights the distribution of buildings based on their GHG emissions levels, categorized as "Very Low," "Low," "Medium," and "High."
 - b. Key Observation: High-emission buildings are concentrated in industrial and older institutional areas. These sectors face operational challenges requiring customized energy solutions.
 - c. Policy Implication: High-emission properties should be prioritized for retrofitting and renewable energy initiatives to reduce their environmental impact.
2. Electricity Usage Across Buildings:
 - a. The electricity usage map provides a granular view of building consumption levels, categorized similarly to GHG emissions.
 - b. Key Observation: Areas with high electricity usage overlap significantly with high GHG emission zones, indicating a direct correlation.

- c. Policy Implication: Addressing energy efficiency in high-usage buildings can lead to substantial reductions in both electricity demand and GHG emissions.
3. Spatial Analysis:
 - a. Both maps reveal clear patterns of compliance and inefficiency, with central Boston buildings showing better compliance rates but still contributing significantly to GHG emissions and electricity usage.

Recommendations:

1. Target High-Emission Buildings:
 - a. Focus resources on buildings with "High" GHG emissions for tailored retrofitting programs and renewable energy adoption.
2. Promote Energy Efficiency:
 - a. Encourage high electricity-usage buildings to adopt energy-saving technologies, particularly in sectors like manufacturing and healthcare.
3. Improve Compliance in Non-Urban Areas:
 - a. Increase outreach efforts in suburban neighborhoods with lower compliance rates to ensure city-wide adherence to BERDO standards.

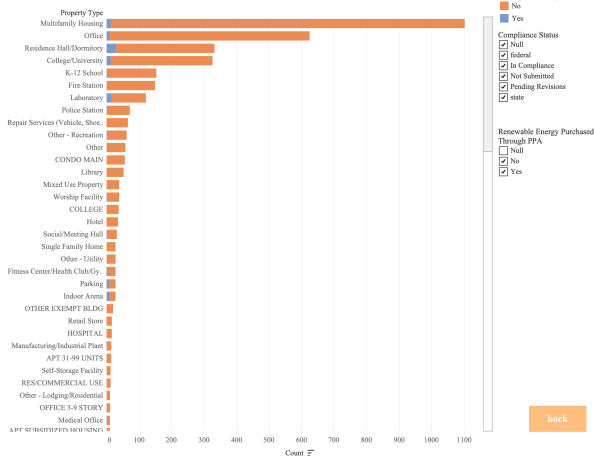
Property Analysis

Property Analysis: As a director of advocacy, I would like to understand the attributes that contribute to BERDO compliance to define initiatives to improve program success.

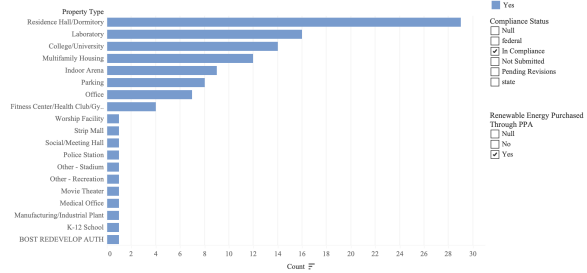
Power Purchase Agreement (PPA) Participation and Compliance

- *Observations:* Properties participating in renewable energy PPAs are significantly more likely to achieve compliance with BERDO standards. However, many compliant properties did not purchase renewable energy through a PPA, indicating that PPA participation itself is not a strong influence on compliance.
- *Insight:* Promoting PPA adoption can facilitate compliance since renewable energy use directly supports sustainable building practices and aligns with BERDO's emission reduction goals.

BERDO Compliance by Property Type and PPA Participation



BERDO Compliance by Property Type and PPA Participation

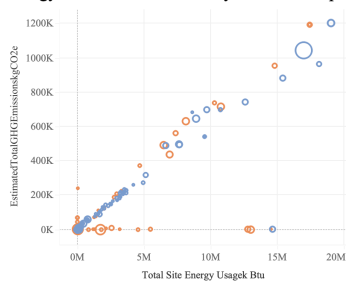


Energy Consumption and GHG Emissions Correlation

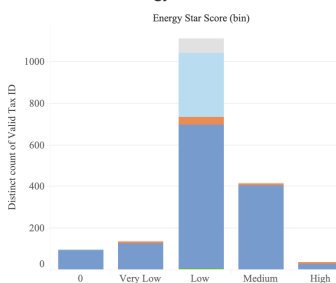
- Scatterplot Analysis:** A direct positive correlation exists between energy usage and GHG emissions. The largest emitters per square foot include hospitals and recreation facilities while manufacturing/industrial buildings demonstrate the highest energy consumption.
- Energy Star Score Distribution:** Most properties fall within the "Low" or "Very Low" Energy Star score categories, underscoring the need for interventions to improve energy efficiency.
- Insight:** High operational demands in sectors like healthcare result in substantial energy usage and emissions, even for energy-efficient buildings. Tailored interventions for such property types are critical.

Energy and Emissions Insights by Property Type

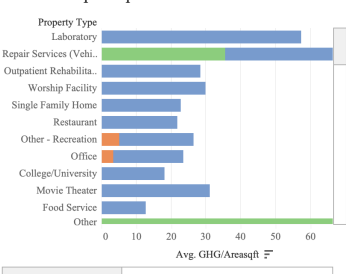
Energy and Emissions Intensity's Relationship



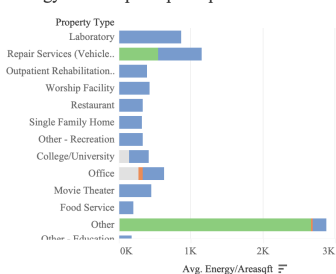
Distribution of Energy Star Score



Emissions per Sq. Ft.



Energy Consumption per Sq. Ft



BERDO Report Status

7 - Accepted by BERDO ..

6 - Rejected by BERDO ..

1 - No Data Reported

0 - Not Claimed

Compliance Status

6 - Rejected by BERDO ..

7 - Accepted by BERDO ..

Property Type

Adult Education

Ambulatory Surgical Cent..

Automobile Dealership

Bar/Nightclub

College/University

Courthouse

Distribution Center

Enclosed Mall

Energy/Power Station

Financial Office

Fire Station

Fitness Center/Health Clu..

Food Service

Hospital (General Medica..

Hotel

Ice/Curling Rink

Indoor Arena

K-12 School

Laboratory

Library

Manufacturing/Industrial ..

Medical Office

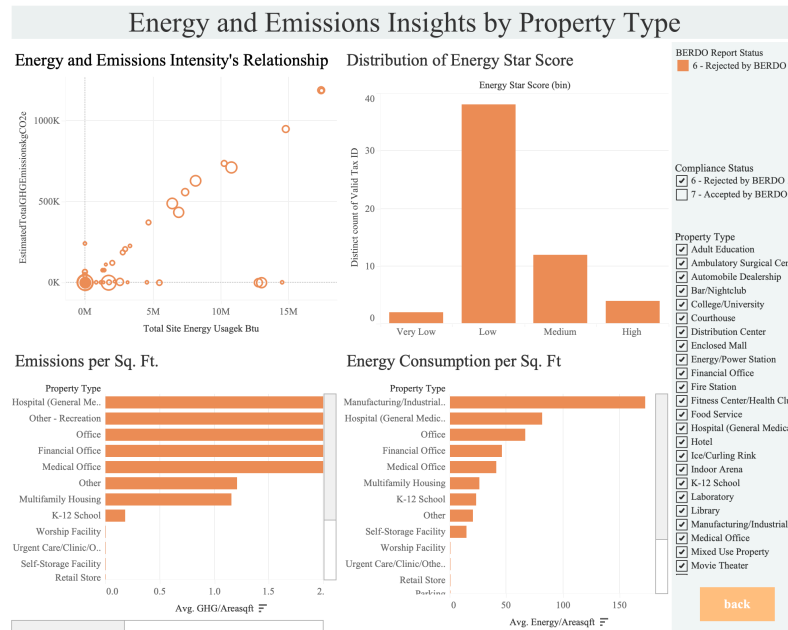
Mixed Use Property

Movie Theater

back

Compliance Challenges for Hospitals

- Despite achieving high Energy Star scores, hospitals face compliance rejections, likely due to their high energy and emissions intensities.
- *Insight:* Addressing these challenges requires policies recognizing operational necessities while promoting innovative energy solutions specific to high-demand property types.



BERDO's Effectiveness in Reducing GHG Emissions:

The Year-to-Year Comparison visualization highlights a significant decrease in greenhouse gas (GHG) emissions from 2022 to 2023. This decline strongly indicates that BERDO is making measurable progress toward its goal of reducing building-related emissions. Key initiatives, such as the adoption of renewable energy through Power Purchase Agreements (PPAs) and improvements in energy efficiency, likely contributed to this progress.

Sustained Impact

The sharp drop in emissions suggests that compliance measures are beginning to take effect. However, further analysis is necessary to determine whether:

- This reduction represents a sustained trend across multiple years or a one-time occurrence.
- The decrease in emissions is uniformly distributed across building types or if it is concentrated in specific sectors.

Energy Efficiency Improvements

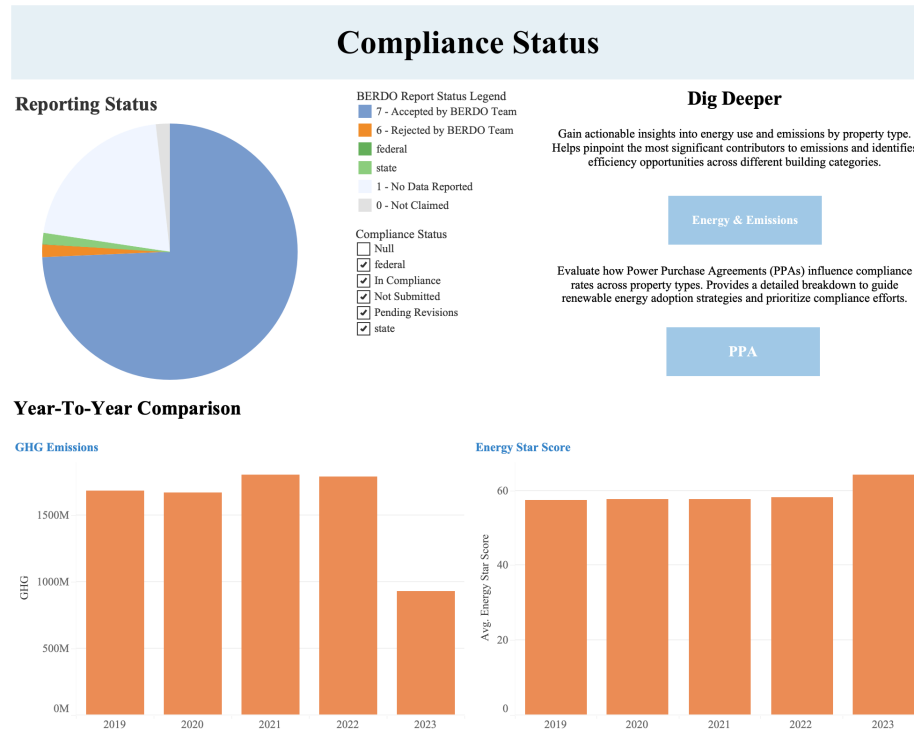
The upward trend in the Energy Star Score over the same period provides additional evidence of BERDO's effectiveness. This improvement reflects the increased adoption of energy-efficient practices and technologies, further reinforcing the ordinance's impact.

Future Focus

While the reduction in GHG emissions is promising, ongoing efforts are needed to sustain and amplify this progress. Additional analysis and targeted interventions should focus on:

1. Identifying and replicating successful compliance strategies in high-emission sectors.
2. Ensuring equitable distribution of emissions reductions across all property types.

These findings underscore BERDO's role in driving Boston closer to its goal of net-zero emissions by 2050 and emphasize the importance of reinforcing successful initiatives to maintain momentum.



Policy Recommendation:

Promote Renewable Energy Adoption

- Expand incentives for Power Purchase Agreements (PPAs) and other renewable energy programs to encourage wider participation. The data shows a strong correlation between PPA participation and compliance, indicating this as a key area for driving further emissions reductions.

Support Energy-Intensive Sectors

- Develop tailored programs and interventions for energy-intensive sectors like hospitals and manufacturing buildings, which face unique operational demands. Policies should balance these demands with achievable compliance measures, such as offering technical support for energy efficiency upgrades or incentivizing innovative energy solutions.

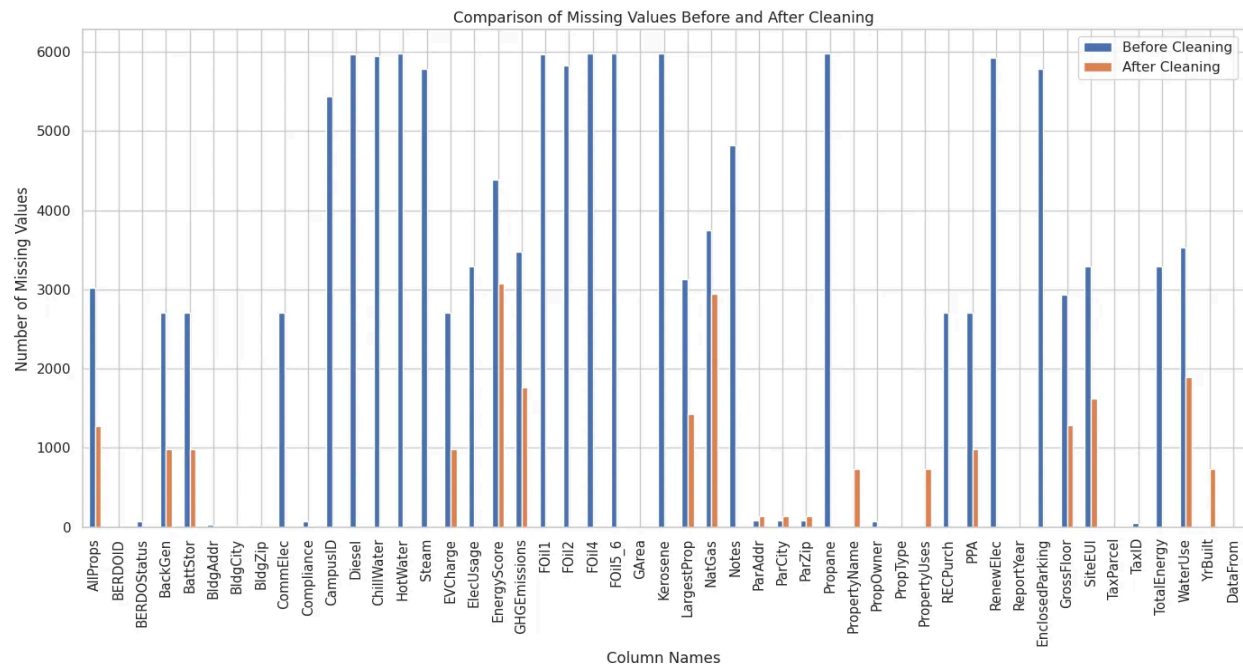
Data Quality for BERDO Database

Project Overview: As the Director of Advocacy, I am focused on enhancing the BERDO database to bolster our city's environmental policy-making. Our project aims to identify and rectify data quality issues to ensure accurate, consistent, and comprehensive data collection, which is vital for supporting effective sustainability initiatives.

Objectives: The key objectives of this initiative include:

1. Enhancing the accuracy of data entries through validation and correction.
2. Ensuring consistency across the dataset for reliable data analysis.
3. Improving data completeness by addressing gaps and eliminating unreliable metrics.

Progress Summary:



1) Data Accuracy Improvements:

- a) Completed Actions: Key fields such as 'TaxParcel' and 'TaxParcelID' have been validated and corrected to ensure accurate data integration. Data types across various fields have been standardized to facilitate uniform analysis.
- b) Impact: These actions have minimized data discrepancies, thereby enhancing the reliability of the dataset.

2) Data Consistency Enhancements:

- a) Completed Actions: Standardization efforts across the dataset have ensured uniform data entries and format alignments, improving the integrity of data joins and comparisons.
- b) Impact: The consistency in data has enabled smoother integration and more robust data-driven decision-making.

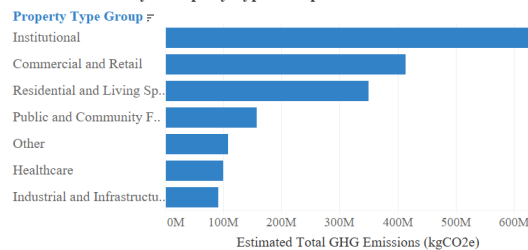
3) Completeness of Data:

- a) Completed Actions: We systematically identified and removed rows and columns with substantial missing or unreliable data to focus our analysis on more complete metrics.
- b) Impact: This has streamlined the dataset, enhancing the quality and relevance of the insights derived from the data.

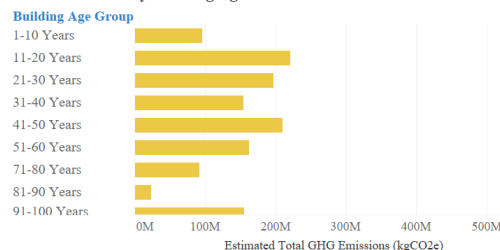
Visualization Summary: The Tableau dashboard provides a detailed analysis of GHG emissions categorized by property type, building age, and neighborhood:

GHG Emissions Analysis by Categories

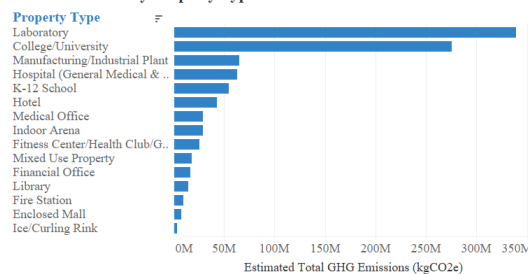
GHG Emissions by 7 Property Type Groups



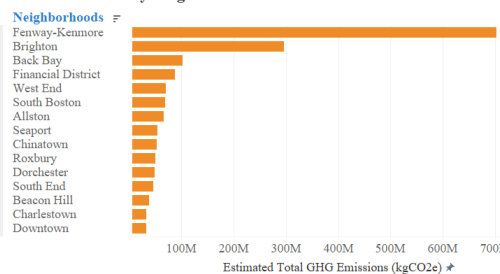
GHG Emissions by Building Age



GHG Emissions by Property Type



GHG Emissions by Neighborhood



Dynamic Filter

- ☒ (All)
- ☒ Adult Education
- ☒ Ambulatory S...
- ☒ Automobile De...
- ☒ Bar/Nightclub
- ☒ College/Univer...
- ☒ Courthouse
- ☒ Distribution C...
- ☒ Enclosed Mall
- ☒ Energy/Power ...
- ☒ Financial Office
- ☒ Fire Station
- ☒ Fitness Center...
- ☒ Food Service
- ☒ Hospital (Gen...
- ☒ Hotel
- ☒ Ice/Curling Rink
- ☒ Indoor Arena
- ☒ K-12 School
- ☒ Laboratory
- ☒ Library
- ☒ Manufacturin...
- ☒ Medical Office
- ☒ Mixed Use Pro...
- ☒ Movie Theater
- ☒ Multifamily H...
- ☒ Museum
- ☒ Non-Refrigera...
- ☒ Office
- ☒ Other
- ☒ Other - Educat...

1. GHG Emissions by Property Type Group showcases emissions from different sectors such as Institutional and Commercial, highlighting key contributors.
2. GHG Emissions by Building Age categorizes buildings by age, illustrating higher emissions in older structures and underscoring the need for targeted retrofitting.
3. GHG Emissions by Neighborhood identifies specific areas with elevated emissions, supporting the development of localized reduction strategies.
4. Dynamic Filters significantly enhance the dashboard's interactivity and user-friendliness, allowing stakeholders to tailor the visualization to focus on specific data subsets. This slicer tool not only aids in precise policy formulation but also greatly improves the interpretability of the data, making the dashboard a highly effective tool for detailed and focused analysis.

Conclusion: Our enhancements to the BERDO database have markedly improved data quality, providing a solid foundation for our advocacy efforts. The integration of detailed visual analytics, including dynamic filters, further supports our capability to effectively analyze GHG emissions and aid in formulating informed environmental policies and strategies for Boston. Moving forward, we will continue to refine these improvements and expand our analytical capabilities to support the city's sustainability goals.

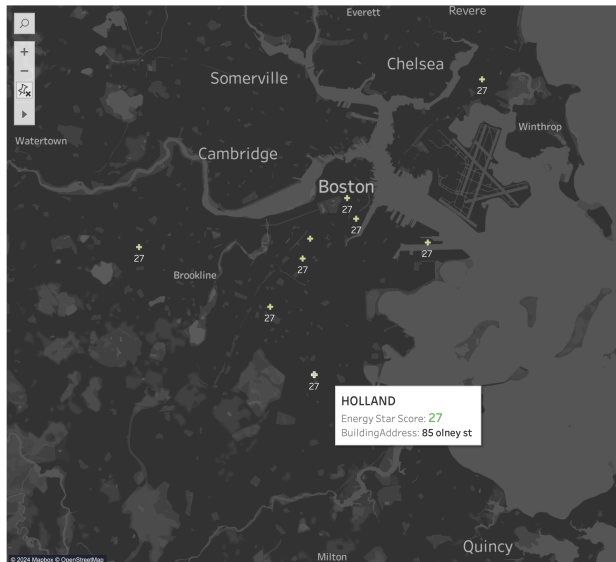
Property Scorecard

Project Overview: As a director of advocacy, I would like an emissions scorecard (using 2025 standards, Energy Star score) for BERDO-required buildings to target specific recommendations for deficient buildings.

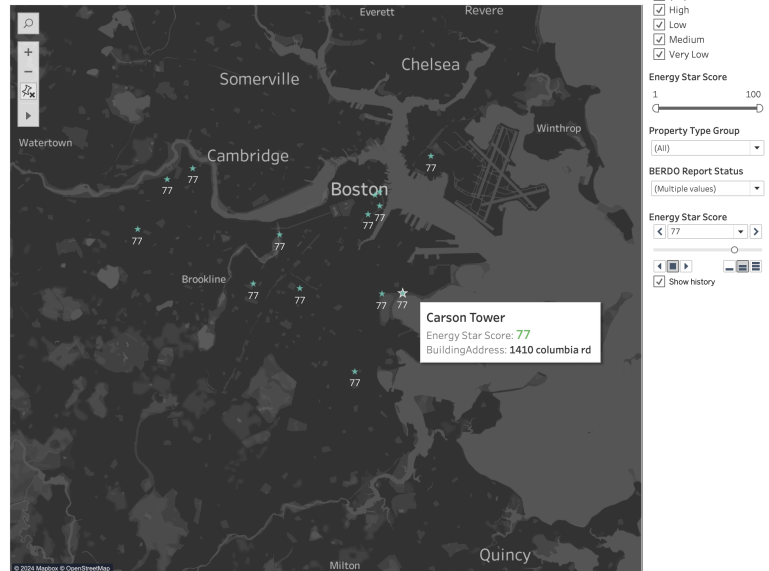
Identifying Deficient Properties

Filtering low and very low Energy Star Scores pinpoints properties with critical energy inefficiencies. For example, Holland (27 Energy Star Score) is a clear candidate for immediate retrofitting.

Property Energy Star Score Mapping



Property Energy Star Score Mapping



Sector-Specific Recommendations

Filtering by property type ensures that recommendations are tailored to the unique challenges of each building sector. For instance:

- Public and community buildings can implement solar panels and energy-efficient lighting systems.
- Industrial buildings might need updated equipment or better energy monitoring.

Compliance Enforcement

Using BERDO Report Status identifies gaps in compliance, such as buildings that are “Not Claimed” or rejected. This allows advocacy to focus on engaging these property owners.

Geographical Insights

Mapping enables a spatial analysis of energy performance. For instance, clustering of high-performing buildings in certain neighborhoods might reveal successful local initiatives, while scattered low-performing buildings indicate areas needing additional support.